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Organizational Learning Processes and Outcomes: Major Findings and Future Research Directions

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Abstract. We trace the evolution of research on organizational learning. As organizations acquire experience, their performance typically improves at a decreasing rate. Although this learning-curve pattern is found in many industries, organizations vary in the rate at which they learn. In order to understand this variation, we separate organizational learning into four processes: search, knowledge creation, knowledge retention, and knowledge transfer. Within each process, we present research on how dimensions of experience and of the organizational context affect learning processes and outcomes. Our goals are to describe major findings and to identify opportunities for future research. The article concludes with a discussion of research directions that are likely to be productive in the future. These directions include investigating how new technological and organizational developments are likely to affect organizational learning.

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Introduction

Large increases in productivity typically occur as organizations gain experience in production. This phenomenon is referred to as a learning curve, an experience curve, a progress curve, and learning by doing. Learning curves are found to characterize not only the production of a wide range of products, including trucks (Epple et al. 1996), semiconductors (Hatch and Mowery 1998), hardware components (Egelman et al. 2017), and chemicals (Lieberman 1984), but also the launch of new products (Gopal et al. 2013) and the establishment of new manufacturing facilities (Salomon and Martin 2008). Learning curves are also demonstrated in service- and knowledge-intensive industries, such as medical care (Pisano et al. 2001, Reagans et al. 2005, Bhargava and Mishra 2014, Ibanez et al. 2017), software development (Boh et al. 2007), and biotech (Jain 2013). The productivity gains associated with learning by doing are typically very large (Levitt et al. 2013).

Research on organizational learning accelerated in the early 1990s. Because the 1980s were a time of great concern about productivity in the United States (Krugman 1991), understanding sources of productivity gains—such as learning—received increasing attention. We focus on research published since 1990 in our review. *Management Science* has published much of this research, which was conducted by researchers in different disciplines, including organizational theory, operations management, strategic management, economics, and engineering. Our focus is on empirical research. We include key results from other sources and other methods, such as computational models, when they advance understanding of organizational learning. Our intent in reviewing the literature is not to be exhaustive, but rather to identify current themes, major findings, and opportunities for future research.

We begin with a historical overview of research on organizational learning. We then present major findings and identify current research themes. Our review is organized according to the learning processes of search, knowledge creation, knowledge retention, and knowledge transfer. Within each section, we discuss how dimensions of experience and of the organizational context affect the learning processes and outcomes as well as research opportunities. We conclude with a discussion of promising research directions for the future. These directions include understanding how new technological and organizational developments, such as machine learning and business platforms, are likely to affect organizational learning.

Historical Overview of Research

Psychologists discovered learning curves at the individual level more than 100 years ago. Focusing on the behavior of individuals, psychologists found that the more individuals practiced a task, the less time they took and the fewer errors they made (Thorndike 1898, Thurstone 1919, Ebbinghaus 1964). Shapira (2020) provides an overview of research on individual learning from these early studies to the current day.

Although early work on individual learning focused on reinforcement learning from direct experience, Bandura (1971) argues that most learning by individuals occurs through exposure to "social models"—examples of other individuals. As Bandura (1971) notes, relying on differential reinforcement of trial-and-error behavior to shape desired responses is not feasible in areas in which mistakes are costly and/or tasks are complex. Social learning can occur through observation of other individuals performing the task or through instruction about how to perform the task. Social learning has been an active research area with a particular focus on learning from "peers" in educational settings (see Epple and Romano 2011 for a review). Researchers have also examined how social learning can occur in commercial organizations. Exploiting a department store's pseudo-random assignment of staff, Chan et al. (2014) analyzes how peers affect the productivity of individual salespeople in a store's cosmetics department. Each brand of cosmetics operates a different counter and hires its own salespeople, which permits the investigation of learning from peers at the same counter and learning from peers at different counters as well as learning from the salesperson's own direct experience. Although all three forms of learning are significant predictors of individual sales revenue, learning from high ability peers at the same counter is more important than learning from peers at adjacent counters or learning from one's own direct experience.

Although social learning by individuals can contribute to organizational learning, organizational learning is more than the aggregation of individual learning curves. Most organizations have some division of labor, which results in individuals performing different tasks. The performance of certain tasks (e.g., in engineering) could contribute more to learning at the organizational level than the performance of other tasks. Further, because members of organizations are interdependent, learning how to coordinate members' activities is a critical part of organizational learning. Consistent with the importance of learning to coordinate, Reagans et al. (2005) find that individual, team, and organizational experience all contribute to the learning curves of surgical teams. Individuals get better at performing their particular tasks as they gain experience with the procedure. As team members gain experience working together, they learn how to coordinate their interdependent activities. As the hospital performs more procedures, the time to perform each procedure decreases. The last effect could reflect social learning between teams, a topic that we return to in our discussion of knowledge transfer.

Wright (1936) publishes the first evidence of a learning curve at the organizational level of analysis. He demonstrates that the amount of labor required to produce an aircraft decreases at a decreasing rate as the organization produces more aircraft. Much of the research on organizational learning curves between the publication of Wright's (1936) classic article and the 1980s focuses on investigating the functional form of the relationship between the unit cost of production and experience measured by cumulative output (Yelle 1979) and on extending the analysis of learning to different products. Learning curves are found to characterize the production of both discrete products, such as aircraft (Alchian 1963), and products made by continuous flow processes, such as petroleum (Hirschmann 1964). Using a production function approach, Rapping (1965) shows that productivity gains associated with experience are not a result of greater inputs of labor or capital or to economies of scale. Although early research on learning curves focuses on manufacturing industries, researchers are increasingly studying learning in service settings, especially hospitals (Kelsey et al. 1984, Pisano et al. 2001, Reagans et al. 2005).

Learning curve analysis is used as a tool for managing the operations of organizations as well as for strategic decision making. On the operational side, learning curves can be used for planning (e.g., production schedules, workforce assignments, training, material orders, delivery commitments, budgeting, technology implementations) and monitoring and improving performance (e.g., Levy 1965, Dolinsky et al. 1990, Kantor and Zangwill 1991, Gaimon 1997, Carrillo and Gaimon 2000, Arlotto et al. 2013). Strategically, firms can use learning curves to predict competitors' costs (Henderson 1984), to decide whether to enter a market, and to determine pricing strategy (Raman and Chatterjee 1995, Balachander and Srinivasan 1998). One strategy based on the learning curve is described by Conley (1970), who argues that the firm that produces the most units would have the lowest cost and the greatest profits. That is, firms are advised to build production volume in order to decrease costs and increase profits.

Evidence began to accumulate that learning was more complicated than the relationship between a performance metric and experience. Several high-profile firms failed to follow their expected learning curves.

Learning curves are often characterized in terms of a progress ratio, which is defined as the reduction in unit costs associated with each doubling of cumulative output. For example, an 80% progress ratio implies that with each doubling of cumulative output, costs decline to 80% of their previous value. Douglas Aircraft did not achieve an 80% progress ratio on its production of the DC-9 (Leonard and Pilarski 2018), which rendered Douglas receptive to an offer from McDonnell. The two firms merged to form McDonnell-Douglas in 1967. Another example of unit costs not following the expected learning curve is Lockheed's production of the L10-11, Tristar, during the 1970s and early 1980s (Argote and Epple 1990). Although Lockheed's costs initially followed a learning curve as cumulative output increased, after production slowed, costs rose and remained higher than the level achieved before the slowdown for the rest of the production program (Benkard 2000). The Lockheed case suggests that the knowledge acquired from learning might not be cumulative, as the classic learning curve implied, but rather decay or depreciate.

A review reveals considerable variation in the rate at which organizations learn. Dutton and Thomas (1984) plot a histogram of the progress ratio found in more than 100 production programs in manufacturing organizations in different industries (see Balasubramanian and Lieberman 2010 for a similar analysis at a later date). Although all but one firm evidences a reduction in unit costs with experience, the extent of reduction varies widely, ranging from the very rapid progress ratio of 55%, which indicates that costs declined to 55% of their previous value with each doubling of cumulative output, to a progress ratio of 96%, which indicates only a 4% reduction. The modal progress ratio was 81% to 82%. Further, Dutton and Thomas (1984) note that there is often more variation in progress ratios between organizations within the same industry than between industries. Several subsequent empirical studies find evidence of variation in learning rates across plants within the same organization producing the same product (Hayes and Clark 1986, Argote and Epple 1990, Chew et al. 1990). The variation across units of an organization suggests that knowledge transfer is not automatic and can be challenging to achieve. If knowledge transfer were easy to achieve, the performance of poorly performing organizational units should converge to the performance of the better performing units.

Examining the processes of organizational learning provides insights into why the strategy of building the most units in order to achieve the lowest costs and the greatest profits might not be effective. First, if knowledge transfers or "spills over" across firms in an industry, the advantages of increasing production volume are less than when the transfer does not occur because other firms benefit from the knowledge a focal firm acquires. Cho et al. (1998) argue that knowledge transfer helps explain how Samsung, a Korean firm that produced semiconductors, was able to beat competitors in the production of dynamic random-access memory (DRAM) even though its levels of cumulative output were less than those of competitors in the United States and Japan.

Second, if knowledge depreciates, the benefits of cumulative production are less than when knowledge persists through time. Knowledge depreciation could have contributed to Samsung's ability to achieve preeminence in a semiconductor product, DRAM, even though its cumulative output levels were less than those of firms in the United States and Japan. When depreciation occurs and recent output is more important than cumulative output, a new entrant to the industry would not be at a competitive disadvantage if its recent output were comparable to the recent output of competing firms.

Research on organizational learning occurred in parallel to research on organizational learning curves until the 1980s. Two streams exist within the organizational learning research. One stream, most associated with Argyris, focuses on the defensive routines members of organizations often invoke that prevent learning (Argyris 1990). Another stream, most associated with March (Cyert and March 1963), emphasizes how organizations encode lessons from experience into routines that guide future performance. Levitt and March (1988) published an influential theory piece in this latter tradition.

These streams of organizational learning research and research on organizational learning curves comingle to some extent in the 1990s. The organizational learning research provides insights into the processes through which learning occurs and, thus, is relevant to understanding the variation observed in learning rates in firms. The organizational learning curve provides a framework for analyzing the effect of various processes on performance. Although work on organizational learning up to the 1980s relies mainly on case studies or simulations, later work tests theory on empirical data collected from the field and accounts for issues such as endogeneity that can arise in naturalistic data (Miner and Mezias 1996).

The Organizational Learning Cycle

Dimensions of Experience

Learning begins with experience. By experience, we mean a unit of task performance (e.g., performing a surgical procedure). The organization interprets the experience to create knowledge. The interpretation of experience can be challenging and subject to biases, such as "superstitious learning," a situation in which "the subjective experience of learning is compelling, but the connections between actions and outcomes are misspecified" (Levitt and March 1988, p. 325).

Researchers characterize experience along different dimensions because different types of experience are found to have different effects on organizational learning (Argote et al. 2003). Perhaps the most fundamental dimension of experience is whether the experience is acquired directly by the focal unit or indirectly from the experience of another unit. Learning from others is discussed in the section on knowledge transfer. We focus on dimensions of experience that have received attention in the management literature: how novel the experience is, whether the unit of task performance is considered to be a success or failure, how ambiguous the experience is, the temporal dimensions of experience (timing and pace), and its heterogeneity.

Organizational Learning Processes

Organizational learning is a process through which experience performing a task is converted into knowledge, which, in turn, changes the organization and affects its future performance. For analytic purposes, we parse the overall learning process into processes of search, knowledge creation, knowledge retention, and knowledge transfer. The latter three processes are suggested by Argote (2013). To those processes, we add the process of search. Our manuscript focuses on learning within organizations as well as learning by organizations. We include several studies that focus on learning by individuals in organizational contexts because those studies explicate the mechanisms through which organizational learning occurs. Throughout the manuscript, the term "organization" broadly refers to the units within an organization (e.g., groups, departments, etc.) as well as the organization itself.

The learning processes are interrelated. We illustrate their interrelationship by considering the learning processes from the perspective of a focal organizational unit. *Creating knowledge* is at the core of organizational learning. An organizational unit can create knowledge from its own experience or vicariously from the experience of other units. For analytic purposes, we discuss studies that focus on learning from an organization's own experience under knowledge creation and studies on learning from others' experience in the knowledge transfer section. *Search* processes are intertwined with the creation and transfer of knowledge. Search, which can be internal or external to the focal organization, is aimed at discovering alternatives and their consequences (March and Simon 1958, Cyert and March 1963).

The organizational unit's interpretation of experience results in knowledge. The knowledge can be tacit (Polanyi 1966) or explicit. *Retaining knowledge* involves embedding it in a repository, such as a routine or a transactive memory system. Knowledge in the various repositories can affect the future performance of the organization. For example, with experience, an organizational unit develops knowledge of who knows what. Because members know who to consult to solve problems, task performance becomes faster. Thus, experience leads to the knowledge of who knows what, which changes the organizational unit and improves future performance.

Once a focal organization has acquired and retained knowledge, it can *transfer* it to other units within the organization and/or to other organizations. In addition, knowledge can unintentionally spill over from the focal organization to other organizations in the environment. Transferring knowledge can result in the creation of new knowledge in the unit providing the knowledge as well as in recipient units (Gruenfeld et al. 2000, Miller et al. 2007).

The Organizational Context

The context is the set of interrelated conditions that form the backdrop for organizational learning. The context includes both characteristics of the environment external to the focal organization, such as its competitiveness or degree of regulation, as well as internal characteristics of the organization, such as its structure, culture, and identity. The context interacts with experience to affect learning (Argote and Miron-Spektor 2011). Some contexts facilitate learning, whereas others impede it.

Argote and Miron-Spektor (2011) differentiate between (a) the *active context* and (b) the *latent context*. The active context includes the basic elements of organizations—members and tools—that interact with the organization's task to influence the interpretation of experience. The latent context, which includes design features of the organization, affects who are members of the organization, what tools they have, and which tasks they perform. A key difference between the active and latent contexts is that the active context is capable of action, whereas the latent context is not.

Although individual members are the medium through which most learning occurs in organizations, the capabilities of tools in the form of machines to learn have increased significantly in recent years. Whereas individuals have a general intelligence factor that facilitates performance on a wide range of cognitive tasks, Brynjolfsson and Mitchell (2017) note that machines do not possess such a general artificial intelligence. Currently, machine learning is most suited for performing particular types of tasks, such as tasks involving learning a function that relates well-defined inputs (e.g., an image in a patient's medical record) to well-defined outputs (e.g., a diagnosis); tasks in which the function linking inputs to outputs does not change rapidly over time; tasks in which large "training" data sets linking inputs to outputs can be created; tasks with clear goals, performance metrics, and unambiguous feedback; tasks that do not involve a long chain of logic or depend on taken-for-granted assumptions; and tasks that do not require explanations of how decisions were made. In addition to the limits in machines' capabilities to perform certain tasks, whether machine learning is adopted also depends on social, organizational, strategic, economic, and legal factors. Thus, humans are likely to continue to play a major role in organizational learning because they are more effective than machines "at many tasks, especially those that require creative reasoning, nonroutine dexterity and interpersonal empathy" (National Academies of Science, Engineering, and Medicine 2017, p. 3).

Knowledge acquired through learning by doing resides in several retention bins or repositories in organizations (Walsh and Ungson 1991, Argote and Ingram 2000). The knowledge repositories include individual employees, the organization's routines and processes, its tools, its culture, and its transactive memory system of who knows what. Knowledge acquired from learning can be embedded in individual employees, including managers or leaders, engineers and technical support staff, and direct production workers. In order for learning to be organizational, the knowledge an individual acquires would have to be embedded in a supra-individual repository, such as a routine, so that the knowledge would persist in the organization even if the individual were to depart.

Knowledge can be embedded in the organization's routines (Cyert and March 1963, Nelson and Winter 1982). A routine is a repetitive pattern of interdependent tasks performed by multiple members of an organization (Feldman and Pentland 2003). For example, as it acquires experience, a hospital emergency unit develops routines for treating patients that embed knowledge about best treatment practices. The routines enable the organization to perform faster and more reliably (Cohen and Bacdayan 1994).

Knowledge can also be embedded in the organization's tools and culture. For example, as an automotive assembly plant acquires experience, the plant fine-tunes its hardware and software to produce higher quality products. As members acquire experience working together, they develop a shared language or set of common terms (Weber and Camerer 2003)—an important aspect of the organization's culture that enables members to perform tasks faster and more reliably.

The organization also develops a transactive memory (Wegner 1986) as it acquires experience. A transactive memory system (TMS) is knowledge of who knows what and who is best at what. More formally, a TMS is a collective system for encoding, storing, and retrieving information (Lewis and Herndon 2011), which develops from experience working together (Liang et al. 1995) and improves task performance (Ren and Argote 2011). Task assignment is improved because members know who is good at which tasks. Problem solving is also enhanced because members know who to consult for advice.

We organize our review by the organizational learning processes. Within each process, we first present research on how dimensions of experience affect organizational learning. We then discuss how the organizational context affects learning. We analyze how characteristics of the organization's members (e.g., their diversity or stability) and tools affect and are affected by organizational learning. Next, we discuss how the following contextual features affect organizational learning: the organization's design (e.g., whether it is a specialist or a generalist and its structure, incentives, physical layout, and training programs), the organization's culture and norms, its absorptive capacity, its slack resources, its power distribution, and social networks within and across organizations. An article could appear in more than one place in our review because, for example, it examines more than one process or analyzes both the effects of dimensions of experience and the effects of contextual conditions.

Search

Organizational search, the process of seeking solutions to existing or anticipated problems, can result in improving existing organizational routines or capabilities or developing new ones (Cyert and March 1963, Nelson and Winter 1982, Dosi and Nelson 1994). In their behavioral theory of the firm, Cyert and March (1963) introduce the concept of problemistic search, which is stimulated by a problem and aimed at finding a solution. A problem occurs when an organization either fails to achieve at least one of its goals or anticipates such a failure. Goals take the form of aspirations, which are a function of the organization's own previous experience and the experience of "comparable" organizations. Problemistic search occurs when an organization's performance falls below an aspiration level, and thus, alternatives to current activities are sought. Slack search, on the other hand, occurs when an organization is performing well and has excess resources that allow for experimentation (Levinthal and March 1981). Slack search often occurs far from existing routines and focuses on experimenting with new alternatives that cannot be justified in terms of their expected returns in the short term but could lead to the development of new capabilities in the long run (Levinthal and March 1981). Refinements of the theory have occurred (see Posen et al. 2018 for a review on problemistic search).

Organizational search is also described in terms of whether the activity contributes to refining existing knowledge or to developing new knowledge. In an influential piece on organizational learning, March (1991) introduces the concepts of exploitation and exploration. Exploitation is related to activities such as "refinement, choice, production, efficiency, selection, implementation, (and) execution," whereas exploration is related to activities such as "search, variation, risk taking, experimentation, play, flexibility, discovery, (and) innovation" (p. 71). March concludes that organizations need to balance exploitation and exploration activities, which spurred considerable research on the topic (e.g., Gibson and Birkinshaw 2004, He and Wong 2004, Andriopoulos and Lewis 2009).

Finally, organizational search is described as having various dimensions. For example, Katila and Ahuja (2002) argue that a firm's search efforts vary across two dimensions: search depth, which refers to how frequently a firm reuses its existing knowledge, and search scope, which is related to how widely a firm explores new knowledge. Focusing only on exploration activities, Rosenkopf and Nerkar (2001) argue that search efforts can occur within a firm or technological boundary or can cross boundaries and categorize search into four types: local, internal boundary spanning, external boundary spanning, and radical search.

Dimensions of Experience

Novelty of Experience. Novel experiences are argued to influence search behaviors. For example, Denrell and March (2001) describe a "hot stove" effect, arguing that poor outcomes on a novel decision could lead to avoiding similar choices in the future and redirecting search in directions that might not be optimal. On the contrary, good performance on novel decisions are shown to reinforce these choices and limit an organization's scope of search to exploitation (Rhee and Kim 2014). Notably, Eggers and Suh (2019) find that responses to negative performance feedback on novel experiences are contingent on an organization's prior domain experience. In the context of U.S. mutual fund companies, firms cease exploration and increase exploitation when they experience negative feedback on new funds launched in new domains and increase both exploitation and exploration when they experience negative feedback on new funds launched in experienced domains.

Success vs. Failure Experience. Performance feedback theory (Greve 2003) suggests that organizations typically regard experience as a success if the performance is higher than the organization's aspiration levels and a failure otherwise (see Greve and Gaba 2020 for a recent review of research on performance feedback). Aspirations are a function of the organization's own previous experience and the experience of other organizations (see Beckman and Lee 2020 for a recent discussion of how these other organizations or social referents are chosen). Researchers have theorized that the intensity and direction of search depend on the extent to which organizations interpret an experience as a success or failure. In addition, even *anticipated* future successes or failures can lead to the search for new solutions (Bhardwaj et al. 2006).

It is shown that organizational performance far above aspiration levels leads to more distant search in the form of slack search (e.g., Levinthal and March 1981, Miller and Chen 2004, Baum et al. 2005). However, repeated successes are argued to lead organizations to limit their search for new knowledge (Nelson and Winter 1982, Song et al. 2003). This effect of repeated success on search is found at the individual level as well, at which it is shown that inventors who have succeeded in the past limit their search space to neighborhoods close to their existing knowledge (Audia and Goncalo 2007). In addition, organizations can fall into "competency traps," in which an alternative with known performance properties is preferred over one with uncertain properties (Levitt and March 1988). In this case, organizations might continue to exploit an inferior procedure that led to successful outcomes instead of exploring a potentially superior procedure.

In contrast, when firms are experiencing failures or performing below aspirations, they are likely to engage in problemistic search. Problemistic search initially starts out in the form of local search and refinements in existing knowledge and gradually develops into distant search and explorative search when the initial search efforts are ineffective (e.g., Levinthal and March 1993, Greve 1998, Miller and Chen 2004, Baum et al. 2005, Billinger et al. 2014). For example, Baum and Dahlin (2007) find that performance near aspiration levels triggers local search and exploitation, whereas performance distant from aspiration levels (both far below and above) fosters nonlocal search and exploration. However, when organizations perform extremely poorly, they can actually decrease their level of search (Chen and Miller 2007, Joseph et al. 2016), consistent with the "threat-rigidity" hypothesis (Staw et al. 1981). Desai (2008) finds that, upon experiencing failures, organizations with less operating experience and poor legitimacy refrain from engaging in search activities that involve risk.

Although the literature suggests that performance both below and above aspirations can increase search efforts, the types of risk involved in search activities might differ under the two situations because organizations have different motivations to search. For example, Xu et al. (2019) find that low-performing firms are more likely to engage in deviant risk-taking behavior (e.g., bribery) to find short-term solutions than high-performing firms. On the other hand, high-performing firms engage in more aspirational risk-taking behavior (e.g., R&D) to sustain long-term competitive advantage than low-performing firms.

Ambiguity of Experience. An experience can be causally ambiguous when the relationship between inputs and outputs of a task is not clear. Ambiguous experiences are theorized to affect search processes. For example, Rahmandad (2008) show in a simulation that, if the performance feedback from a search activity is delayed and thereby ambiguous, it could cause organizations to abandon such search activity although there could be positive organizational returns to sustaining search in the long run.

Pace of Experience. There is some evidence that the pace of experience, the rate at which experience occurs, affects search behaviors. For example, Zellmer-Bruhn (2003) finds that interruptive events that temporarily prevent the organization from accumulating experience in its normal routines could trigger active cognitive processing that leads to the search for new and better routines. Hayward (2002) similarly finds, in a study on merger-and-acquisition experiences of firms, that too little time between experiences could limit organizational search capabilities for learning. Interestingly, he also finds that too much time between experiences could dampen organizational search as well.

In its extreme form, experience can occur very infrequently. Such rare experiences influence search behaviors as well. Treating an organizational crisis as a rare event, Rerup (2009) finds qualitative evidence that such an event can trigger a search for solutions to organizational problems that were not obvious prior to those events. His case study shows that the search for solutions occurs both within (such as reexamining the current organizational structure) and outside of the organization (such as becoming more attentive to the external environment). Christianson et al. (2009) similarly find that rare (and disastrous) events could lead organizations to search for solutions in areas that they would not have considered otherwise.

Heterogeneity of Experience. The heterogeneity of experience is found to influence search behaviors. For example, Haunschild and Sullivan (2002) find that experiencing a variety of failure experiences leads to a broader search for solutions to organizational problems than experiencing homogeneous failure. Similarly, Kim et al. (2009) find that accumulating both success and recovery experiences leads to the search for better solutions compared with accumulating only success experiences or only recovery experiences. Finally, Zollo (2009) finds that accumulating a stock of heterogeneous experience could prevent organizations from engaging in superstitious learning.

Organizational Context

Members. Individual members' characteristics can influence their tendency to explore versus exploit (Laureiro-Martínez et al. 2010). For example, whether organizational decision makers have career experience in particular business functions (e.g., research and development, marketing, finance, legal, operations) or obtained advanced science-related degrees influences their tendency to search for new solutions (Barker and Mueller 2002). In addition, high-ability employees who find it easier to meet minimal performance requirements under a fixed-salary incentive system engage in more exploration than low-ability employees, presumably to acquire new knowledge and skills using their slack time (Lee and Meyer-Doyle 2017). Finally, Lee (2019) finds that individuals with higher organizational tenure are more capable of learning the knowledge and skills of newly colocated organizational peers and utilizing such knowledge and skills to increase exploration than those with lower organizational tenure.

Member diversity and stability is also found to influence search behaviors because they affect the knowledge and experience base of organizational members. Taylor and Greve (2006) find that teams with members with diverse knowledge and experience produce innovative but high-variance outcomes, presumably because they engage in more exploratory search. Similarly, March (1991) shows in a simulation that turnover of organizational members can facilitate exploration. Interestingly, Franke et al. (2013) suggests that effective exploratory search can be achieved by bringing in individuals who have expertise in contextually distant domains that share an analogous organizational problem.

Tools. The use of tools can also influence search behaviors. Using a simulation model, Kane and Alavi (2007) demonstrate that tools, such as knowledge repositories, portals, and virtual team rooms, facilitate exploitative search, whereas electronic communities of practice facilitate explorative search. Based on an analytic model, Lee and Van den Steen (2010) conclude that organizations benefit more from a knowledge management system when they are large, experience the same issues repeatedly, have high turnover, and encounter problems about which there is less general knowledge. Kim et al. (2016) find that a knowledge management system implemented in a retail grocery chain assists managers who have few alternative sources of knowledge to search effectively for solutions to improve store performance. On the other hand, Haas and Hansen (2005) find in a consulting organization that downloading documents from a knowledge repository is associated with poor team performance, especially for experienced teams.

Specialist vs. Generalist Organizations. Specialist organizations tend to engage in search that is localized to a particular domain, whereas generalist organizations tend to search more broadly across a variety of domains. Specialists also are likely to search in more depth than generalists (Kang and Snell 2009).

Organizational Structure. Organizational structure is theorized to influence organizational search. Using simulation models, studies have shown that decentralization enables organizations to explore new solutions and thereby prevents them from prematurely converging on suboptimal solutions (Siggelkow and Levinthal 2003, Ethiraj and Levinthal 2004, Siggelkow and Rivkin 2005). In another simulation study, Csaszar (2013) presents results suggesting that reducing hierarchy could promote exploration. Finally, in terms of balancing between exploitation and exploration, Fang et al. (2010) find that organizations divided into semi-isolated subgroups are better at maintaining the balance between exploitative and exploratory search than isolated subgroups or randomly connected individuals.

Empirical studies have yielded results largely consistent with simulation models. For example, Jansen et al. (2006) show that decentralization increases explorative innovation in organizations. Interestingly, in the context of mergers and acquisitions, Puranam et al. (2006) show that the immediate structural integration of firms harms the acquired firm's innovation performance when the acquired firm is in the stage of its innovation trajectory in which exploration is more important than exploitation. However, when the acquired firm is in the stage in which exploitation is more important than exploration, structural integration leads to more innovations. Perretti and Negro (2006) find that either having a flat hierarchy or a hierarchy with middle managers who can effectively coordinate interdependencies between different organizational projects leads to exploratory search. Jansen et al. (2009) find that structurally differentiated organizations can balance exploitative and exploratory search by adopting integration mechanisms, such as senior team social integration and cross-functional interfaces.

A growing stream of research shows that organizational structure moderates the relationship between performance feedback and organizational search. For example, Rhee et al. (2019) find that, in hierarchical business groups with multiple subunits, problemistic search at the subunit level is significantly intensified when group-level managers are cognitively aware of the problems at the subunit level and provide support for solving such problems. In another study in the context of Indian firms, Vissa et al. (2010) find that business group–affiliated firms and unaffiliated firms respond differently to performance feedback. Business group–affiliated firms are more likely to increase search when they are underperforming in terms of market position compared with unaffiliated firms because business group–affiliated firms' aspiration levels are more externally oriented than those of unaffiliated firms. In addition, Joseph et al. (2016) find that centralized organizational structures amplify responses to performance feedback above aspirations but attenuate them below aspirations.

Ownership structures also influence organizational search behaviors. For example, Wu (2012) shows that, when organizations go public through initial public offerings, they decrease search that explores new and recently developed knowledge but increase search building on scientific knowledge. In addition, O'Brien and David (2014) show that different types of ownership leads to different intensities of search behaviors in response to positive performance feedback.

Incentives. Incentive designs are shown to influence search behaviors. For example, Lee and Meyer-Doyle (2017) show that switching individuals' incentives from a pay-for-performance system to a fixed-salary system promotes exploratory search. Ederer and Manso (2013) find that an incentive system that tolerates early failure and rewards long-term success is most effective for exploratory search. Using a simulation model, Baumann and Stieglitz (2014) suggest that higher-powered incentives could lead to excessive competition among organizational individuals and thus dampen search efforts.

Physical Space. The design of organizational physical space is also found to influence search behaviors (Allen 1977). For example, Lee (2019) finds that reconfiguring seating arrangements so that previously separated individuals are seated closer together leads to learning that stimulates exploratory search and also improves the effectiveness of exploitative search. Catalini (2018) shows that colocation between organizational members decreases search costs and promotes novel collaborations. Clement and Puranam (2017) similarly show in a simulation model that physical separation mandated by a formal organizational structure could limit organizational individuals' search for novel social interactions.

Organizational Culture and Norms. Organizational culture influences the pattern of search behaviors of individuals within organizations. Gambeta et al. (2019) find that strong firm–employee relationships lead to employees engaging in more exploitation and less exploration. In a study of inventors in the hard disk–drive industry, Audia and Goncalo (2007) find that successful individuals working in firms with a norm for exploration (i.e., distant search) are less likely to generate incremental ideas than successful individuals working in firms without such norms.

Absorptive Capacity. Absorptive capacity is an organization's "ability to recognize the value of new information, assimilate it, and apply it to commercial ends" (Cohen and Levinthal 1990, p. 128). Absorptive capacity is gained by accumulating experience, and the higher an organization's absorptive capacity, the more likely it is that it effectively searches for novel solutions externally (Cohen and Levinthal 1990, Rothaermel and Alexandre 2009). For example, in a study on the acquisition behaviors of Ontario nursing home chains, Baum et al. (2000) find that organizations generally search for new solutions in search spaces in which they have prior experience and in spaces in which other large or similar organizations are actively searching. Several comprehensive reviews on the topic of absorptive capacity have been published (see Zahra and George 2002, Lane et al. 2006, Todorova and Durisin 2007, Volberda et al. 2010).

Slack Resources. Slack resources are characterized as both motivators and moderators of search processes. Several factors, including organizational performance above aspiration and environmental conditions, can affect the development of slack resources in organizations (Sharfman et al. 1988). A search process that is stimulated by slack resources is called a "slack search" (Cyert and March 1963, Levinthal and March 1981). Slack search is commonly characterized as exploratory and occurring in search spaces distant from those related to existing organizational routines and capabilities (Levinthal and March 1981). For example, Iyer and Miller (2008) find that firms with more slack engage in more acquisitions, an activity that involves risk and distant search. Slack resources are also found to increase the intensity of search. For example, Chen (2008) find that firms' financial slack increases the R&D intensity of firms. Slack in terms of human resources is also found to influence organizational search. For instance, Lecuona and Reitzig (2014) find that having slack in human resources who possess tacit and firm-specific knowledge facilitates the search for novel solutions when the firm faces competitive pressures in its external environment.

Power and Status. Bunderson and Reagans (2011) suggest that power and status differences between individuals within an organization can lead individuals with lower power or status to engage in less risk taking and experimentation, activities that are core to exploratory search. Loch et al. (2013) find that power and status differences within a group impair problem solving because members rely too much on high-status members. Lower-status members in organizations typically have lower levels of psychological safety compared with higher-status members (e.g., Edmondson 2002, Nembhard and Edmondson 2006), which presumably can contribute to the limited search activities of lower-status members. In addition, Singh et al. (2010) find that individuals with low status in organizations typically have poor connectivity with other organizational individuals, which limits their search for new information.

Social Networks. Social networks open up opportunities for search. Hansen (1999) finds that weak interunit ties in organizations help project teams search for valuable knowledge in other subunits, whereas strong ties are necessary to transfer tacit knowledge. In addition, organizational alliances are suggested to facilitate distant search (Rosenkopf and Almeida 2003). Moreover, using a simulation model, Lovejoy and Sinha (2010) show that maximizing the number of parallel conversations per period while encouraging individuals to dynamically churn through a large set of conversational partners fosters the creation of ties that facilitate the search for new solutions. Rogan and Mors (2014) find that the network density and network contact heterogeneity of managers affect their capability to effectively balance exploratory and exploitative search.

Interestingly, not all network ties are useful for effective search. Borgatti and Cross (2003) suggest that the decision to seek information from others depends on relational characteristics between the information seeker and the information source. When the seeker knows the source's expertise, positively evaluates such expertise, and perceives that the seeker has access to the information source, the seeker is more willing to seek information from that source. Similarly, Singh et al. (2010) find that individuals with higher status, higher tenure, and better connectedness are able to take advantage of network structures more when searching for information than individuals lacking those characteristics. Nerkar and Paruchuri (2005) argue that inventors reduce the cost of search by signals of quality based on their colleagues' positions in networks within the firm. When individuals are central in the communication network and span structural holes (Burt 1992), it is more likely that their knowledge is selected by another inventor.

Further, there is some evidence that centrality and structural holes positively reinforce each other.

Research Opportunities

Although the effect of success versus failure experience has received considerable research attention, the other dimensions of experience have not been studied much in relation to search. It would be useful to deepen our understanding of the relationship between different dimensions of experience and search. For example, how would organizations shift their search behaviors when faced with ambiguous experience? Would organizations engage in local search to interpret the ambiguous experience using their own expertise or engage in distant search to make sense of the experience using knowledge from new domains? In addition, ambiguous experience can potentially facilitate organizations' exploration. Ambiguous performance feedback from exploration activities might allow organizations to continue to search more distantly instead of ceasing exploration activities when faced with initial negative outcomes despite the potential of long-term positive outcomes. Further, the relationship between heterogeneous task experience and search merits additional attention. Could moderate task heterogeneity benefit organizational search by encouraging the balance between exploitation and exploration?

Dimensions of the organizational context that warrant more attention in relation to search are (1) characteristics of members, (2) whether the organizations are specialists or generalists, and (3) routines. Specifically, what are the mechanisms driving the different search behaviors of specialists and generalists? Do specialists tend to search more locally because they lack motivation (e.g., less necessary to acquire knowledge outside of their core domains), abilities (e.g., low absorptive capacity in other knowledge domains), or opportunities (e.g., the lack of social networks conducive to acquiring new knowledge)? Further, it would be interesting to consider the role of routines in balancing exploitation and exploration. For example, would it be possible for organizations to develop routines that foster engaging in both exploitation and exploration? Also, how do newcomer socialization processes and organizational norms influence the relationship between member turnover and search behavior? Although newcomers can bring new expertise and encourage organizations to search in new domains, if conformity pressure is high, newcomers might instead search extensively in organizations' existing knowledge domains to conform to the new setting.

Knowledge Creation

As organizations perform a task, they acquire knowledge. Members learn how to perform their individual tasks better and learn who is good at which tasks. Tools are calibrated. Routines are refined and structures are fine-tuned. Members interpret whether task performance was a success or failure, which can stimulate search. Thus, performing a task can result in the creation of new knowledge in the organization. We focus in this section on learning from an organization's own direct experience.

Dimensions of Experience

Novelty of Experience. Research suggests that organizations could develop a greater amount of knowledge by balancing experience high and low in novelty rather than focusing on one type of experience (March 1991). For example, He and Wong (2004) find that pursuing high levels of both exploitative and explorative innovation strategies increases manufacturing firms' sales growth rate by improving firms' processes and products. Similarly, Katila and Ahuja (2002) show that firms introduce more new products when they balance exploitation and exploration.

Success vs. Failure Experience. Several studies investigate whether organizations learn differently from their own success and failure experience. Research finds that organizations learn from their own failures (Chuang and Baum 2003, Madsen 2009). For example, Stan and Vermeulen (2013) show that fertility clinics that have a higher chance to experience failures because they admit complex cases learn at a faster rate than clinics that admit only relatively simple cases. Complex cases enable the clinics to deepen their understanding of knowledge domains, explore new solutions, and develop tools and routines to capture and transfer the knowledge gained from the complex cases. In a study of accidents in U.S. railroads, Baum and Dahlin (2007) find that organizations learn from their own failures when the organizations' performance (i.e., accident rates) does not deviate far from their aspiration levels.

Several studies focus on identifying the conditions that facilitate organizations' learning from their own failures. Haunschild and Sullivan (2002) show that specialist airlines learn more from failures with heterogeneous causes than from failures with homogeneous causes. As noted earlier, it is likely that failures with heterogeneous causes trigger organizations to conduct broader search for solutions than failures with homogeneous causes. Studying recalls of automakers, Haunschild and Rhee (2004) find that experience with voluntary—but not involuntary—recalls reduces subsequent involuntary recalls. The researchers suggest that learning from voluntary recalls is much deeper than learning from involuntary recalls, underscoring the role of volition in learning. In the context of hospitals in which surgeons perform heart surgeries, Desai (2015) finds that organizations learn less from failures when their failures are relatively concentrated in origin, meaning that failures are incurred mostly by a particular organizational unit or a specific member, compared with when failures are more broadly dispersed across units or individuals. Finally, prior studies suggest that organizations learn more effectively from unexpected or salient failure experiences than from expected or less salient failure experiences (e.g., Madsen 2009, Rerup 2009, Madsen and Desai 2010, Ocasio et al. 2020).

Studies that directly compare the effectiveness of organizations' learning from their own successes and own failures provide mixed findings. Madsen and Desai (2010) find that organizations learn more effectively from failures than successes. The researchers suggest that failures help organizations identify gaps in organizational knowledge and increase organizations' motivation to fill the knowledge gap. On the other hand, KC et al. (2013) find that individuals within organizations learn from their own successes but not from their own failures. This pattern is suggested to be due to individuals attributing their own successes to internal factors from which they can draw lessons while attributing their own failures to external factors that would not provide generalizable lessons. Kim et al. (2009) find that organizations learn from both successes and failures only after accumulating a certain threshold of success and failure experience. Further, as noted earlier, success and failure experience enhances learning from the other type of experience because contrasting different types of experience enables organizations to generate more useful lessons.

On the topic of learning from failures, recent developments have occurred on the methodological front. Bennett and Snyder (2017) propose a revised method to examine learning from failures, in which they recommend measuring failure experience within a moving time window instead of across an entire sample period and not including success experience in the same model with failure experience.

Ambiguity of Experience. Causally ambiguous experience can hinder learning from an organization's own experience as it is difficult for organizations to interpret such experience (Levitt and March 1988). A delay between an action and an outcome could make an experience causally ambiguous. Diehl and Sterman (1995) find that teams perform more poorly as the delay in feedback becomes longer. Similarly, Repenning and Sterman (2002) show that the delay in the feedback of a process improvement initiative hinders employees from understanding the effectiveness of the initiative accurately.

Timing of Experience. Several studies find that organizations learn not only *by* doing but also *before* and *after* doing. Pisano (1994) finds that learning before doing, such as conducting research prior to actual production, is beneficial only when an industry already has a deep knowledge base. In contrast, learning by doing is more useful than learning before doing when the knowledge base is not well developed. Similarly, Eisenhardt and Tabrizi (1995) show that the computer industry, characterized by an evolving knowledge base, does not benefit from planning before production. Focusing on after-event reviews, Ellis and Davidi (2005) show that the learning effect is greater when groups are debriefed on both successes and failures than when groups are only debriefed on failures. Furthermore, Morris and Moore (2000) find that individuals could learn by reflecting on how they could have done better after an experience (upward counterfactual comparisons).

One study focuses on the life cycle of organizations. Aranda et al. (2017) suggests that mature organizations learn more from their own past experience than from other organizations' experiences because mature organizations have sufficient levels of their own experience from which to draw lessons.

Pace of Experience. The rate at which organizations acquire experience can affect their processes of learning from their own experience. For example, acquiring experience at an uneven rate can hinder learning from experience (Argote et al. 1990). Furthermore, although learning from infrequently occurring or rare experience can be challenging, such rare experience can provide organizations with valuable lessons (e.g., Lampel et al. 2009). Focusing on acquisition events, Zollo (2009) suggests that learning from rare experience with ambiguous performance metrics is particularly difficult because organizations are likely to interpret the outcomes as successes and engage in superstitious learning. Rerup (2009) shows that organizations have to invest deliberately to learn lessons from rare experience. Madsen (2009) finds that coal miners learn from both frequently occurring minor accidents and rarely occurring disasters but through different mechanisms. Minor accidents contribute to learning by alarming employees to comply with organizations' safety routines, whereas rare disasters trigger fundamental changes in the organization's safety routines. Similarly, Christianson et al. (2009) find that rare events provide organizations with the opportunity to improve existing routines.

Heterogeneity of Experience. Research on the effects of heterogeneous experience on the creation of knowledge yields mixed results. Several studies suggest that heterogeneous task experience could hinder organizational learning and performance. Fisher and Ittner (1999) find that product variety negatively influences auto plants' productivity. Clark et al. (2018) show that the organizational learning rate diminishes when the goals of heterogeneous tasks are not aligned. Gopal et al. (2013) find that the deleterious effect of manufacturing new heterogeneous products is mitigated when the plant has past experience in producing products that are similar to the new product. On the other hand, other studies suggest that heterogeneous experience facilitates organizations' learning from their own experience (e.g., Schilling et al. 2003, Wiersma 2007, Narayanan et al. 2009, Egelman et al. 2017). Underscoring the benefits of heterogeneous task experience, Staats and Gino (2012) find that individuals learn how to effectively switch between different tasks as they gain more heterogeneous task experience. In addition, Jeppesen and Lakhani (2010) find that individual problem-solver's experiences in technological or social domains different from the focal problem facilitate developing successful solutions in science problem-solving contests.

These divergent findings on experience heterogeneity can be reconciled by considering the level of analysis at which learning occurs and by considering the degree of heterogeneity. Boh et al. (2007) find that the effect of heterogeneous task experience differs depending on the levels of analysis: individuals benefit the most from specialized task experience, whereas groups benefit the most from working on heterogeneous tasks that were related. Schilling et al. (2003) find in a laboratory study that moderate task heterogeneity is beneficial and suggest that performing tasks that are different but related allows groups to develop a more abstract and deeper understanding of knowledge applicable to different tasks. Egelman et al. (2017) find that the performance of a contract manufacturer is improved when the organization produces multiple generations of the same product family and that knowledge transfer between related products is the mechanism driving the benefit of related product variety. Similarly, in the context of hospitals, Clark and Huckman (2012) find that cardiovascular patient care quality is improved when hospitals also specialize in areas related to cardiovascular care. The authors suggest that the cospecialization enables cardiovascular specialists to access new knowledge and insights held by specialists of other care units. Thus, the performance of related tasks benefits the performance of a focal task, suggesting that some heterogeneity in experience can be beneficial for task performance.

Organizational Context

Members. Member diversity is found to affect organizational learning. For example, in the semiconductor industry, Macher and Mowery (2003) find that team diversity moderates the relationship between experience and organizational performance such that functionally diverse teams learn more from their experience than functionally homogeneous teams.

Several studies examine the effect of team stability on organizational learning. On the one hand, team stability is found to have a positive effect on organizational learning. For example, Reagans et al. (2005) find that team stability (i.e., the average number of times team members worked together) has a positive effect on the performance of surgical teams. Similarly, Huckman et al. (2009) find that team stability positively contributes to the performance of software teams. In the context of a software support services operation, Narayanan et al. (2009) find that new member entry into a workgroup reduces productivity. They also find that productivity suffers when the variety of experience in a workgroup is reduced because of a member exiting. On the other hand, in the context of delivering mail, Wiersma (2007) finds that employing a modest level of temporary employees increases organizational learning rates because of the new knowledge that temporary employees bring into the organization.

Tools. Tools can positively influence learning by facilitating the acquisition, storage, and sharing of information. Research on the effect of tools on organizational learning has focused mainly on information technology or knowledge management systems. For example, Ashworth et al. (2004) find that the adoption of an information system in a bank increases learning from both direct and indirect experience.

Specialist vs. Generalist Organizations. Evidence on whether generalist or specialist organizations learn more effectively is somewhat mixed. Most studies find that specialist organizations learn more from experience than generalist organizations (Ingram and Baum 1997, Haunschild and Sullivan 2002). For instance, in the context of retail banks, Barnett et al. (1994) find that specialist banks have higher returns on assets as a function of experience than generalist banks. In addition, Ingram and Baum (1997) find that organizations that operate over a large geographical area (i.e., "geographical generalists") learn less from their own experience than organizations that specialize in a smaller number of areas. Interestingly, in the context of airlines learning to increase customer satisfaction, Lapré and Tsikriktsis (2006) find that the best specialist airline (i.e., focused airline) learns faster than the best generalist airline (i.e., full-service airline) although there is no difference in learning rates, on average, between the two types of airlines. In the context of hospitals, KC and Terwiesch (2011) examine the effect of the degree of specialization on performance at different levels of analysis. After controlling for the potential effect of selectively admitting easy-to-treat patients, firm-level specialization (e.g., the fraction of cardiac-related admissions among all admissions) does not have a positive effect on patient care quality, whereas operating unit-level specialization (e.g., the fraction of patients with a particular cardiac illness among all cardiac patients) does improve patient care quality. An exception to the pattern of specialist organizations learning more from experience than generalist organizations is provided by Haunschild and Rhee (2004), who find that generalist automobile manufacturers learn more from voluntary product recalls than specialists do.

Organizational Structure. Bunderson and Boumgarden (2010) find that, in self-managed teams dealing with stable tasks, more team structure (i.e., higher degrees of specialization, hierarchy, formalization) facilitates learning. Stan and Puranam (2017) show that clinics with integrator structures (e.g., coordinator roles) recover more quickly from a regulatory shock that shifts the interdependence patterns among employees and have steeper learning rates after the shock than clinics lacking integrator structures. Ben-Menahem et al. (2016) find that both formal coordination structures and informal coordination practices contribute to team-based knowledge creation. Lee (2019) finds that organization members become better at both creating new knowledge and refining existing knowledge when their seats are moved closer to the seats of individuals from whom they were previously separated. Bresman and Zellmer-Bruhn (2013) find that team- and organizational-level structures have different effects on learning: greater team-level structure is associated with increased internal and external team learning, whereas a greater organizational-level structure is associated with decreased internal and external team learning. Sorenson (2003) finds that vertical integration hinders the rate of learning in firms in stable environments and facilitates learning in volatile environments.

Incentives. Lee and Meyer-Doyle (2017) find that switching from a pay-for-performance incentive system to a fixed-salary system promotes learning because individuals engage in more exploration efforts under the latter system. Similarly, Lee and Puranam (2017) find that switching from a pay-for-performance incentive system to a fixed-salary system promotes learning through increased knowledge sharing and collaboration between individuals. As noted earlier, Clark et al. (2018) find that the learning curve for surgeons' patient care outcomes is influenced by whether the goals of the other incentivized activities are aligned with improving patient care.

Training. The training of individuals or groups within an organization can improve learning (Bell and Kozlowski 2008, Ford 2014). Training is particularly useful when organization members have opportunities to observe experts or experienced members performing tasks (Nadler et al. 2003) because trainees can acquire tacit knowledge through observing experts perform tasks (Nonaka 1991). Studies also show that group training is more beneficial than individual training for collective learning (Liang et al. 1995, Hollingshead 1998). Transactive memory systems—a collective system for encoding, storing, and retrieving information—develop when members are trained in groups. Also characterized as "knowledge of who knows what," transactive memory systems are found to facilitate the creation of knowledge (Gino et al. 2010).

Organizational Culture and Norms. Organizational culture and norms also are found to influence knowledge creation. For example, Edmondson (1999) finds that a culture characterized by "psychological safety" facilitates learning. In psychologically safe environments, members feel safe to express their ideas and take risks (Nembhard and Edmondson 2006, Bunderson and Reagans 2011). When team members emphasize learning in their unit rather than comparing their performance to that of other units, they learn more from their experience (Bunderson and Sutcliffe 2003). Finally, cohesion or liking among group members (Wong 2004) or shared language among organizational members (Weber and Camerer 2003) also facilitates the interpretation of and learning from their own experience.

Absorptive Capacity. Absorptive capacity is determined by the level of prior related knowledge (Cohen and Levinthal 1990), which is accumulated through activities such as research and development (R&D) as well as through training and experience performing the task. R&D is found to facilitate learning not only from external sources but also from an organization's own direct experience. For example, in the context of the chemical processing industry, Lieberman (1984) finds that R&D investments increase the rate of learning among firms. Similarly, Sinclair et al. (2000) find that R&D is associated with the productivity gains observed in a chemical firm.

Slack Resources. Slack resources can open up opportunities for creating new knowledge (Cyert and March 1963). Indeed, several studies find a positive relationship between organizational slack and organizational learning (e.g., Wiersma 2007). However, other studies find an inverted U–shaped relationship between slack resources and new knowledge creation: increases in slack initially facilitate innovation, but too much slack hurts the discipline necessary to produce innovations (Nohria and Gulati 1996).

Power and Status. In a review paper, Bunderson and Reagans (2011) suggest that power and status differences could lead to less collective learning in organizations because such differences hinder experimentation, knowledge sharing, and the development of shared goals. The authors further conclude that these negative effects could be mitigated when individuals with high power and status are collectively oriented and use their power and status for the benefit of the organization. Relatedly, Van der Vegt et al. (2010) find that power asymmetry within teams could hinder team learning, but the effect is contingent on the level of feedback provided: whereas individual-level feedback reinforces the negative effects of power asymmetry on team learning, team-level feedback mitigates those effects.

Social Networks. Social networks are found to facilitate learning and knowledge creation by opening up access to new knowledge sources. As mentioned earlier, Lovejoy and Sinha (2010) show in a simulation model that the ideation stage of innovation projects can be accelerated when individuals dynamically churn through a large set of conversational partners over time. This process creates decentralized networks that facilitate search. The results of empirical studies are consistent with simulation results. For example, Reagans and Zuckerman (2001) find that teams characterized by high demographic heterogeneity have high learning capabilities because the members' networks cut across diverse groups of individuals. Similarly, Tortoriello and Krackhardt (2010) find that ties that span structural holes—ties that bridge otherwise unconnected parts of a network—are helpful for developing new knowledge, especially when individuals who bridge boundaries share common third-party ties.

Research Opportunities

We would benefit from learning more about how organizations learn from their own novel or ambiguous experience. Our understanding of how the pace of experience affects organizational learning would also benefit from additional research. For example, one promising research direction is the relationship between the pace of experience and knowledge creation. On the one hand, experience acquired at an uneven rate could hinder knowledge creation more than experience acquired at a steady rate (Argote et al. 1990). On the other hand, such interruption could give organizations opportunities to reflect on their actions and adopt changes to improve performance in the future (Christianson et al. 2009). It would be interesting to understand the conditions under which organizations could benefit from experience being acquired at an uneven rate.

Of the dimensions of the context that have not been investigated much in relation to knowledge creation, the ones that seem most promising to investigate are routines, tools, and slack resources. Although routines are often thought to be a source of inertia, Guo (2019) hypothesizes and finds that routines can facilitate the creation of knowledge and adaptation to a new task. Although much of the work on tools and organizational learning focuses on the effect of knowledge management systems on search, it would be promising to study the effect of tools such as machine learning on the knowledge creation process. Another interesting direction would be to understand how the distribution of slack resources influences the creation of knowledge. Within organizations, different teams or individuals might have varying levels of access to slack resources held by organizations. Do different organizational members have different perceptions of slack? How would the distribution of slack or the process of negotiating the distribution of slack influence organizational-level knowledge creation?

Knowledge Retention

In addition to the Lockheed case mentioned previously, several other case studies suggest that knowledge acquired through learning might not persist indefinitely (Hirsch 1952, Baloff 1970). Research through the 1980s using the learning curve, however, assumed that learning was cumulative and used cumulative output as the appropriate measure of experience. In this section, we review evidence from studies investigating whether the knowledge acquired from learning is cumulative or whether it depreciates.

Beginning in the 1990s, researchers tested whether learning was cumulative as the classic learning curve implied or whether the knowledge acquired from learning by doing depreciated. For conceptual clarity, we use the term "depreciation" to refer to the decay of knowledge at the organizational level and reserve the term "forgetting" to describe decay at the individual level. Individual forgetting can contribute to knowledge depreciation at the organizational level, but organizational knowledge depreciation is affected by other factors related to the organizational retention bins discussed previously.

Researchers investigated the extent to which knowledge depreciates by estimating a parameter that is a geometric weighting of previous experience. When the parameter is estimated to be less than one, it suggests that knowledge depreciates because past experience receives less weight than recent experience in predicting current production. Studies find evidence of depreciation in different organizational contexts, including shipyards (Argote et al. 1990, Thompson 2007, Kim and Seo 2009), automobile assembly plants

(Levitt et al. 2013), hotels (Ingram and Baum 1997), and fast-food franchises (Darr et al. 1995). Benkard (2000) provides convincing evidence that depreciation occurred in Lockheed's production of the L-1011 TriStar and rules out the explanation that the depreciation was due to changes in the product that made experience from the past obsolete. Comparing depreciation rates across these studies, knowledge depreciates more in contexts with high turnover, such as fast food franchises (Darr et al. 1995), than in contexts with low turnover, such as kibbutzim agriculture (Ingram and Simons 2002). Organizations in which considerable knowledge is embedded in technology (Egelman et al. 2017) exhibit little or no depreciation.

These studies advance understanding by demonstrating that knowledge acquired via learning by doing typically exhibits some depreciation and, thus, is not cumulative as the conventional learning curve implies. Incorporating depreciation into forecasts of future productivity results in better estimates. The studies, however, do not determine the cause of depreciation. Studies that take a fine-grained approach to experience or analyze the organizational context provide insights into potential causes of depreciation.

Dimensions of Experience

Only a few studies examine how dimensions of experience affect knowledge depreciation.

Success vs. Failure Experience. Analyzing accident experience in U.S. coal mining, Madsen (2009) finds that experience from major accidents depreciates at a slower rate than experience from minor accidents. In a study of orbital launches, Madsen and Desai (2010) find that experience from failed launches depreciates less than experience from successful launches.

Pace of Experience. Ramdas et al. (2018) take a fine-grained approach to experience by analyzing surgeons' experience with different medical devices used to perform hip replacements. The researchers find evidence of knowledge depreciation in the use of two particular devices, stems and liners, which are difficult to use. For these devices, as the days since the previous usage increase, the duration of the surgery significantly increases, suggesting that the knowledge about how to use the devices depreciates.

Organizational Context

Agrawal and Muthulingam (2015) compare the extent to which knowledge embedded in three different repositories in the supply chain of a manufacturing facility depreciates. The researchers find that knowledge embedded in technology exhibits the least depreciation over time, knowledge embedded in routines evidences intermediate depreciation, and knowledge embedded in individuals exhibits the most depreciation.

Members. Studies examine the effect of knowledge embedded in individuals on knowledge retention by studying the effect of turnover. David and Brachet (2011) compare the effects of member turnover on the effect of skill decay caused by inactivity or task interference on knowledge depreciation. The contribution of member turnover to knowledge depreciation is found to be about twice the effect of skill decay. Studying the turnover of loan officers, Drexler and Schoar (2014) find that the negative effects of turnover are less when turnover is anticipated than when it is not and when the departing member has time and incentives to transfer knowledge to new members.

Examining the interaction between turnover and the organizational context, Rao and Argote (2006) and Ton and Huckman (2008) find that turnover is less harmful in organizations that rely to a great extent on routines and procedures than in organizations less reliant on procedures. The routines buffer the organizations from the effect of membership change. Similarly, Argote et al. (2018) find that the clear coordination logic of centralized communication networks buffers organizations from the effect of turnover: decentralized networks perform better when membership is stable, whereas centralized networks perform better when turnover occurs. Shaw et al. (2005) find that the departure of members who occupy "structural holes" or bridges between otherwise unconnected individuals is more harmful than the departure of members in dense networks. Kellogg (2011) finds significant evidence of knowledge depreciation in a study of learning between firms in contractual relationships, which is attributed to the loss of relationship-specific capital between firms.

Routines. Given the important role routines play in storing knowledge in organizations, studies examine the extent to which organizations follow routines. Analyzing franchise organizations, Knott (2001) finds that adherence to routines decreases when units leave franchises, which harms unit performance. In their analysis of pharmaceutical firms, Anand et al. (2012) find that adherence to operational routines decays over time, especially when a merger occurs. By contrast, inspections from the Food and Drug Administration and acquisitions appear to halt decay in routines.

Tools. Reinforcing the importance of tools as knowledge repositories, research shows that new members of an organization benefit from the knowledge embedded in tools and routines. Studies of the introduction of a second shift at manufacturing plants find that the new shifts, which are staffed primarily by employees new to the organization, achieve levels of productivity comparable to the existing shifts in record time (Epple et al. 1991, 1996; Levitt et al. 2013). The dramatically faster ramp-ups of productivity on the second relative to the first shifts are due to much of the knowledge acquired by the first shift being embedded in tools and routines that the second shift uses.

Interestingly, although IT systems developed before the advent of machine learning store primarily explicit knowledge, machine learning tools have the potential to embed tacit knowledge that is hard to articulate as well as explicit knowledge (Brynjolfsson and Mitchell 2017). Before recent advances, tacit knowledge that could not be articulated (Polanyi 1966) into a set of rules could not be programmed into a computer. In machine learning, when algorithms are run on training data, the algorithms can capture regularities not even noticed by people. Brynjolfsson and Mitchell (2017) conclude that machine learning algorithms have made it possible to train computer systems to be more accurate than those that are manually programmed for many tasks.

Although the research discussed thus far suggests that knowledge depreciation is bad for the organization because the organization is less productive than it would be if depreciation had not occurred, several researchers suggest that depreciation can have positive effects on the organization (Easterby-Smith and Lyles 2011). Using the term "forgetting" to refer to the loss of an organization's knowledge, de Holan and Phillips (2004) categorize forgetting into accidental and purposeful. Accidental forgetting occurs when an organization does not embed or maintain the knowledge it acquires in organizational memory. By contrast, purposeful forgetting occurs when an organization deliberately fails to embed knowledge or purges knowledge that it deems no longer appropriate or useful. Similar to the benefits of purposeful forgetting, researchers argue that decreasing memory could weaken inertia and increase an organization's ability to adapt (Starbuck 1996). Consistent with this argument, simulations show that the imperfect recall of information from organizational memory can increase exploration and an organization's adaptive potential (Jain and Kogut 2014). Jain (2020) concludes that the productivity-improving properties of organizational memory dominate its inertia-producing properties.

Research Opportunities

Identifying the conditions under which knowledge depreciation has positive versus negative implications for organizations is an important topic that would benefit from additional research. How to determine what knowledge should be purposefully forgotten or purged would also benefit from research. Advocates of forgetting argue that old routines and understandings should be discarded when they are no longer useful or appropriate for a changed environment. Determining when knowledge is no longer useful is a challenging task. Examples exist of knowledge that turned out to be valuable after it was deemed no longer useful. For example, when Steinway decided to put a discontinued piano back into production, it discovered that it did not have records or blueprints of how to make the model (Lenehan 1982).

Very little research has been done on how dimensions of experience affect knowledge retention. Further research is needed to advance our understanding of whether different types of experience depreciate at different rates. For example, heterogeneous task experience has been suggested to foster the development of more high-level, abstract knowledge (Schilling et al. 2003). Would this type of knowledge depreciate at a slower rate than low-level knowledge? Also, would the depreciation rate differ for knowledge gained through exploitation versus exploration? Is the knowledge more likely to be retained when it is acquired before, by, or after doing?

In addition, the topic of contextual influences on knowledge retention merits attention. Research on knowledge retention focuses largely on the role of active contexts (e.g., organizational members, tools), but several interesting questions arise pertaining to the role of latent organizational contexts. For example, how would organizations' absorptive capacity influence the degree of knowledge retention? Because organizations with high absorptive capacity can assimilate valuable information more effectively than organizations with low absorptive capacity (Cohen and Levinthal 1990), would organizations with high absorptive capacity also be more effective in retaining that information than organizations with low absorptive capacity? Another promising topic is the effect of power and status hierarchies on knowledge retention. Would organizations' power or status hierarchies influence the internal process of identifying which knowledge to retain or eliminate from the organization's memory? What about the role of organizational design in knowledge retention?

Knowledge Transfer

Knowledge transfer is the process through which one organizational unit learns from or is affected by the experience of other units (Argote and Ingram 2000). The "unit" could be individuals, groups, or the overall organization. Knowledge transfer is also referred to as vicarious learning or knowledge spillover. The latter term is typically used within the field of economics and connotes unintentional spillover, such as might occur within firms in an industry (e.g., see Irwin and Klenow 1994).

Knowledge transfer can occur through a variety of mechanisms, including hiring employees; reverse engineering products; and acquiring knowledge from suppliers, vendors, consultants, conferences, scientific publications, and patents. Knowledge can transfer through cooperative relationships, such as alliances, joint ventures, and consortia. In general, knowledge transfer occurs by moving the knowledge repositories from one organizational unit to another or by modifying the knowledge repositories of the recipient unit (Argote and Ingram 2000). Carlile and Rebentisch (2003) theorize that knowledge transfer occurs through the processes of retrieving knowledge (e.g., searching for knowledge and assessing its usefulness) and transforming the knowledge (e.g., creating shared language) to be used by other units. Further, Myers (2018) proposes a theoretical model of coactive vicarious learning, in which a learner and a teacher interact to process experience and derive lessons instead of a one-way process of a learner observing and learning from a teacher.

Researchers take several approaches to measure knowledge transfer (Argote and Fahrenkopf 2016). For researchers studying learning curves, just as learning is assessed by the relationship between the experience of a focal unit and the unit's performance, knowledge transfer is assessed by the relationship between the experience of other organizational units and the focal unit's performance. The other organizational units could be units of the same organization as the focal unit (e.g., sister plants producing the same product); units linked to the focal organization through a relationship, such as a franchise or an alliance; or units in firms in the same industry as the focal unit. A statistically significant relationship between the experience of some group of organizations and a focal unit's performance provides evidence of transfer: the focal unit is affected by the experience of other units. Although the relationship could be negative, it is typically nonexistent or positive, which indicates that the focal unit benefits from the experience of the other units. Researchers also measure knowledge transfer through survey questions (Szulanski 1996), through changes in routines (Kane et al. 2005), and through patent citations (Almeida and Kogut 1999, Rosenkopf and Almeida 2003, Alcacer and Gittelman 2006).

The topic of knowledge transfer has received considerable attention in recent years because an organization's ability to successfully transfer knowledge from other organizations or across organizational units can be a source of competitive advantage (Argote and Ingram 2000). Although many studies provide evidence that organizations can transfer knowledge (e.g., Adler 1990, Argote et al. 1990, Salomon and Martin 2008), considerable variance in the extent of knowledge transfer among organizations is found (Szulanski 1996, Lapré and Van Wassenhove 2001). To understand the source of the observed variation in the effectiveness of knowledge transfer, scholars examine factors such as the characteristics of knowledge, knowledge senders, knowledge recipients, and organizational contexts as well as the methods of knowledge transfer such as personnel movement, routine transfer, and so on (for a comprehensive review, see Szulanski and Lee 2020). For example, Zander and Kogut (1995) find that codified knowledge transfers more readily than noncodified knowledge. In this section, we review major findings of how different dimensions of experience and organizational contexts affect knowledge transfer from one organization (or an organizational unit) to another.

Dimensions of Experience

Success vs. Failure Experience. Several studies examine whether organizations could learn from other organizations' failures in order to avoid costly learning by themselves. Research shows that organizations learn from other organizations' failures in various empirical contexts, including railroads (Baum and Dahlin 2007), coal mining (Madsen 2009), and commercial banking industries (Kim and Miner 2007). Recently, several studies have identified conditions that facilitate learning from others' failures, such as the availability of information on others' failures and motivation and ability to learn from others' failures (see Dahlin et al. 2018). For example, Yang et al. (2015) show that Japanese firms entering China learn more from the failures of earlier entrants that have prior network ties with the firms than from the failures of earlier entrants without network ties.

Some research directly compares the effectiveness of learning from others' successes versus failures. Studies generally suggest that organizations learn more effectively from others' failures than from others' successes (Madsen and Desai 2010, KC et al. 2013) for at least two reasons (an exception is Haunschild and Miner (1997), who show that firms learn from both successes and failures of others). First, organizations typically have easier access to information on others' failures than on their successes because organizations' failures are often highly publicized whether by organizations or regulatory agencies (Kim and Miner 2007, Madsen 2009, Madsen and Desai 2010). On the other hand, information on organizations' successes is typically contained internally to sustain their competitive advantage (Katila et al. 2008, Madsen and Desai 2010). Second, organizations are more likely to imitate other organizations' successful strategies blindly without adjusting the strategies to fit their own contexts, whereas other organizations' failures tend to promote deeper search for the fundamental causes of the failures (Sitkin 1992). In a related vein, KC et al. (2013) show that cardiac surgeons learn from other surgeons' failures because the surgeons attribute others' failures to internal factors, such as skills, and search for further learning. On the other hand, the surgeons do not learn from other surgeons' successes because those are attributed to external factors, such as luck, which are not amenable to being imitated. Still, individuals might be able to learn from others' successes if they have easier access to information on others' successes. For example, Song et al. (2018) find that publicly disclosed performance feedback enables employees to identify from whom to learn and to validate the quality of knowledge held by high performers and thereby facilitates the transfer of superior practices from high to low performers.

Ambiguity of Experience. An organization needs to understand the fundamental factors contributing to the observed outcomes of others in order to learn effectively from the experience of others. As expected, research shows that organizations learn more from others' experiences when the underlying knowledge is less ambiguous and, thus, easier to understand than when it is ambiguous (Szulanski 1996, King 1999).

Timing of Experience. When organizations decide to learn from others' experience (e.g., before or after their own task performance) could influence the effectiveness of knowledge transfer. Although some studies show that organizations learn on an ongoing basis from the experience of other units (e.g., Darr et al. 1995), others show that organizations learn from other units' experience accumulated up until the organization's founding but not thereafter (Argote et al. 1990, Baum and Ingram 1998). The latter effect is more likely to be found in contexts such as shipyards and hotels that require extensive investments in physical infrastructure early on that are not amenable to adaptation on an ongoing basis. Aranda et al. (2017) suggest that organizations tend to learn from other organizations earlier in their own life cycle than later because of their own lack of direct experience.

Pace of Experience. The pace of experience is suggested to influence knowledge transfer. For example, as noted earlier, Zellmer-Bruhn (2003) finds that interruptions, such as a change in technologies or restructuring events, triggers teams to evaluate current routines more mindfully and to learn from others if the current routines are unsatisfactory.

Organizational Contexts

Members. Because individual members are important channels to transfer both tacit and explicit knowledge (Argote and Ingram 2000), many studies investigate the effectiveness of transferring knowledge by moving individual members. Although considerable evidence suggests that knowledge transfers to different contexts when individuals move (e.g., Rosenkopf and Almeida 2003, Singh and Agrawal 2011, Jain 2016, Kolympiris et al. 2019), other studies present a more complex picture. For example, several studies suggest that knowledge held by individuals might not be perfectly transferrable across organizations (Huckman and Pisano 2006, Groysberg et al. 2008). Some portion of individuals' knowledge is organization-specific, such as knowledge of who knows what in organizations and familiarity with organizational resources. Such organization-specific knowledge is not likely to transfer to new organizations. For example, members with specialist experience who are required to coordinate more with other organization members than generalists acquire considerable knowledge that is specific to a particular organization and its members. Specialists are found to be less likely to transfer their knowledge to new organizations than generalists (Fahrenkopf et al. 2020). Furthermore, groups utilize the knowledge of new members only when the new members are perceived to have high-status (Bunderson et al. 2013), when the new members provide high-quality knowledge to groups that share a social identity with them (Kane et al. 2005), when the new members have expertise in distinct technical domains (Song et al. 2003), or when the previous work experience of the new members is aligned with the structure of the recipient groups (Fahrenkopf et al. 2020).

Furthermore, the diversity of organizational members can influence knowledge transfer. Several studies suggest that teams with diverse members (e.g., in terms of organizational tenure, geographic locations) are likely to have access to unique knowledge sources outside of the group and thereby transfer unique knowledge to the group (Reagans and Zuckerman 2001, Cummings 2004).

Tools. Tools, including IT systems, such as knowledge management systems, online repositories, and social media, can be effective means to transfer knowledge across geographic, temporal, and organizational boundaries (Wagner et al. 2014, Maslach et al. 2018, Neeley and Leonardi 2018). Despite the potential benefits of moving knowledge through tools, studies identify potential challenges involved in transferring knowledge using IT systems. For example, Hwang et al. (2015) find that, in the early phase of using online knowledge communities, individuals share knowledge only to similar others in terms of hierarchical status and geographical location.

Organizational Structure. Several studies find that hierarchical structures hinder the transfer of knowledge.

For example, middle managers become more reluctant to pass their subordinates' ideas to superiors when the hierarchy is steeper because middle managers do not feel that they have control over the outcomes (Reitzig and Maciejovsky 2015). When organizational units are embedded in a centralized hierarchy, the units do not transfer knowledge directly to each other but rather transfer the knowledge indirectly via corporate headquarters, which could be inefficient (Tsai 2002).

Many studies investigate how an organization's alliance structure (e.g., franchising, shared ownerships, joint ventures, licensing) influences knowledge transfer. Several studies find that knowledge transfers more effectively across units embedded in the same franchise or multiunit chain (Darr et al. 1995, Baum and Ingram 1998). In a related vein, Oxley and Wada (2009) show that a joint venture with a shared ownership structure, compared with a pure contract-based partnership, facilitates the speed and the extent of knowledge transfer across partners. Interestingly, the joint venture structure is also more effective in preventing unintentional knowledge spillover across partners in domains outside of the alliance agreement. Pierce (2012), however, warns that a shared ownership structure could hinder knowledge transfer more than a pure contract-based structure when the shared ownership creates conflicting incentives among the parties involved. Extending our understanding of when the experience of a common owner is most likely to be valuable, Kalnins and Mayer (2004) find that the locality of the experience matters. The experience of local units with a common owner is beneficial, whereas the experience of nonlocal units with a common owner is not. Because incentives are constant across co-owned units, incentives are not a viable explanation of the findings. Communication is a viable explanation because communication is denser for co-owned local units than for co-owned nonlocal units. Reinforcing the importance of communication, Kalnins and Chung (2006) find in a different empirical context that knowledge transfers across high-end hotels owned by members of the same ethnic group.

Incentives. Several studies show that incentives could motivate employees to transfer knowledge. Quigley et al. (2007) show that dyads with a strong knowledge-sharing norm share more knowledge with each other when they have a group-based incentive than when they have an individual-based incentive. Similarly, Lee and Puranam (2017) find that a switch from individual pay-for-performance to a fixed salary incentive system facilitates the transfer of knowledge across employees, most likely by making the collective goal salient.

Physical Space. Several studies show the benefits of colocation on knowledge transfer across units and individuals (e.g., Catalini 2018, Lee 2019). In the context of salespeople in a department store, Chan et al. (2014) show that individuals learn from peers in the same counters through direct observations and active teaching by their peers. Individuals even learn from peers in physically adjacent competing counters by observing their behaviors. Hatch and Mowery (1998) find that semiconductor plants produce fewer defective products upon adopting a new routine when the manufacturing and routine development facilities are colocated than when they are not colocated. The authors suggest that the colocation of a knowledge sender and a recipient facilitates the transfer of tacit knowledge. Gray et al. (2015) presents similar results in the pharmaceutical industry. In addition, the authors examine whether information and communications technologies (ICTs) could help geographically dispersed units to overcome the barriers of tacit knowledge transfer. They find that ICTs are not as effective as face-to-face communication in transferring tacit knowledge. Several studies show that it is difficult to transfer knowledge across geographically dispersed teams because members typically have low levels of a shared understanding (Cramton 2001, Gibson and Gibbs 2006). When teams have psychologically safe climates, teams can overcome the communication barriers and transfer knowledge (Gibson and Gibbs 2006).

Despite the potential benefits of colocation on knowledge transfer, some studies also highlight boundary conditions. For example, in the context of a start-up boot camp, Hasan and Koning (2019) find that individuals are less likely to receive advice from physically proximate others when they already have prior ties with which to interact compared with when individuals do not have such prior ties. In addition, Chatterji et al. (2019) show that the content of the knowledge exchanged among collocated individuals eventually affects firm performance. By contrast, Darr and Kurtzberg (2000) find that strategic similarity is a more important predictor of knowledge transfer than geographic closeness.

Organizational Culture and Norms. Organizational culture and norms are found to affect the extent of knowledge transfer. Both norms that encourage knowledge holders to share knowledge and norms that guide knowledge recipients' use of the received knowledge (e.g., not passing the received knowledge to others without permission) can influence the transfer of knowledge (Merton 1973, Constant et al. 1994, Fauchart and Von Hippel 2008). For example, Di Stefano et al. (2014) find that Italian chefs are more willing to transfer knowledge to recipients who are expected to conform to the norms of knowledge use than to recipients who are less expected to conform. Examining the role of shared social identity on knowledge transfer, Kane et al. (2005) show that groups are more willing to adopt high-quality knowledge communicated by new group members when they share a superordinate social identity than when they do not. Wong (2004) find that groups learn more from individuals outside of the groups when group cohesion is high than low.

In addition, when units within an organization have distinct subcultures, it might be difficult for the units to transfer knowledge across each other as a result of having different languages or work contexts. Bechky (2003) finds that units can overcome this knowledge transfer challenge by paying attention to key differences across units and creating a common ground (e.g., shared language, shared understanding of different work contexts) to communicate with each other.

Absorptive Capacity. Research finds that organizations with high absorptive capacity are more adept at identifying and integrating useful external knowledge and, therefore, learn more effectively from others than organizations with low absorptive capacity (Szulanski 1996, Tsai 2001). Posen and Chen (2013) examine how new entrants, which typically lack absorptive capacity, learn vicariously from incumbent firms. The authors show that problems encountered during learning-by-doing motivate the new firms to search for knowledge from others and facilitate vicarious learning despite the lack of absorptive capacity.

Power and Status. Power and status differentials among organizational members are found to influence knowledge transfer. Focusing on social status arising from social ties, Thomas-Hunt et al. (2003) show that socially isolated members are more likely to share their uniquely held knowledge than socially connected members. Menon et al. (2006) demonstrate that individuals avoid seeking knowledge from rivals within the same organizations because of the threat of losing status and instead seek knowledge from individuals outside their organizations. Raman and Bharadwaj (2012) examine organizational members with power differentials and find that the effectiveness of routine transfer depends on whether the interests of organizations and high-power members are aligned.

Social Networks. Research shows that the characteristics of social relationships among parties influence the effectiveness of knowledge transfer across them. Several studies find that high-quality relationships (e.g., strong, cohesive, embedded ties) facilitate the transfer of knowledge between parties and that the benefit is usually greater when transferring tacit, complex, or private knowledge (e.g., Szulanski 1996, Hansen 1999, Ingram and Simons 2002, Uzzi and Lancaster 2003, Levin and Cross 2004). Levin and Cross (2004) find that trust is an underlying mechanism driving the knowledge transfer advantage of strong ties. In a related vein, units or individuals in a competing relationship within an organization are less likely to seek knowledge from and transfer knowledge to each other (Hansen et al. 2005, Menon et al. 2006). Still, Tsai (2002) shows that frequent social interactions among competing organizational units facilitates knowledge transfer across units. He suggests that having social interaction enables competing units to develop trust and realize opportunities to create synergies.

Many studies examine how the structural characteristics of organizational networks affect knowledge transfer effectiveness. Tsai (2001) finds that relative to organizational units in less central positions, a unit occupying a central position in interunit networks produces more innovations by having access to diverse knowledge developed by other units. This effect is strengthened when the unit has a high absorptive capacity. Also, individuals with higher network range perceive transferring knowledge to be easier because they are familiar with being exposed to and communicating diverse knowledge (Reagans and McEvily 2003). Furthermore, several studies suggest that having shared third parties (e.g., mutual contacts of a knowledge source and a knowledge recipient) facilitates knowledge transfer across the source and the recipient by creating cooperative norms and raising the reputation costs of not sharing knowledge (Tortoriello and Krackhardt 2010, Tortoriello et al. 2012). When the ties of a source and recipient do not overlap, these unshared contacts are referred to as unshared third parties. Consistent with the benefits of shared third parties, Reagans et al. (2015) find that unshared third parties reduce the likelihood of the source and the recipient initiating and sustaining knowledge transfer relationships. Further, the negative effect of having unshared parties decreases as the knowledge possessed by the unshared parties becomes more similar.

Research Opportunities

Considerable research has been done about the effect of success versus failure experience on knowledge transfer. More studies are needed to understand the effects of other types of experience on knowledge transfer. We suggest that our understanding of how different types of experience influence the extent of knowledge *creation* could shed light on the influence of different types of experience on the extent of knowledge *transfer*. For example, organizations might be able to learn more effectively from the heterogeneous outcome experience of others by inferring valuable lessons via contrasting experience as organizations did with their own heterogeneous outcome experience (Kim et al. 2009). Similarly, as suggested by research on a rare experience's influence on knowledge creation (Madsen 2009, Zollo 2009), the rare experiences of other organizations might either facilitate knowledge transfer by drawing attention to the events or hinder knowledge transfer by triggering superstitious learning in which inappropriate inferences are drawn about relationships between the actions of other organizations and observed outcomes. Understanding when knowledge transfer is facilitated or impeded by rare experience is an interesting topic for future research.

Studying the effects of different organizational contexts on knowledge transfer is a promising research direction as well. More research has been done on how characteristics of members, organizational design features, absorptive capacity, power and status, and social networks affect knowledge transfer than on the other contextual dimensions. In terms of organizational norms, it would be interesting to see whether norms of knowledge sharing and norms of knowledge use play complementary or substitutable roles in facilitating knowledge transfer within organizations. For example, would it be enough to have appropriate norms of knowledge use (e.g., ensuring that the shared knowledge would not be used against the original knowledge holder) to encourage knowledge transfer? Another interesting direction would be to examine how different active contexts (e.g., members, tools) complement or substitute each other in facilitating knowledge transfer. For example, suggesting a complementary relationship between members and tools, Galbraith (1990) found that knowledge transfer was greater when individuals were moved along with tools than when only tools were moved. In a similar vein, would a routine transfer be more effective if accompanied by personnel transfer? When would it be most beneficial for organizations to rely on members, and when would it be effective to rely on tools to facilitate knowledge transfer?

Future Directions

In each section on learning processes, we note areas in which there are opportunities for future research on the process. In this section, we discuss future research directions that pertain to the relationships between the different learning processes. We also discuss how new technologies and organizational forms are likely to affect organizational learning. We conclude by discussing methodological developments that have promise for advancing our understanding of organizational learning.

1. A promising question that would benefit from additional research is about the relationship between the processes of organizational learning. Several studies show that learning from direct and indirect experience are complements: the more that organizations learn from their own experience, the more they are able to learn from the experience of other organizations (Bresman 2010). Other studies show that the two learning processes are substitutes: organizations that learn a great deal from their own experience do not learn very much from the experience of others (e.g., Wong 2004). Similarly, examining the relationship between knowledge retention and knowledge transfer, Levine and Prietula (2012) find that the two processes are substitutes: greater access to knowledge in organizational memory decreases the benefits of knowledge transfer. Understanding the conditions under which the learning processes complement or substitute for each other is an important topic that would benefit from additional research.

2. Organizational member stability is found to have different effects on different learning processes. Team member stability has a positive effect on knowledge creation (Reagans et al. 2005) and retention (e.g., Liang et al. 1995). Membership change, however, is found to facilitate knowledge transfer (e.g., Rosenkopf and Almeida 2003, Song et al. 2003, Choi and Thompson 2005, Kane et al. 2005). Understanding how to balance membership stability and membership change and the degree of member stability that is most appropriate for different contexts would advance understanding of organizational learning.

3. A topic that has gained attention recently is the role of physical space in organizational learning. The interactions among individuals is found to be bounded by an organization's physical space even with the development of information and communication technologies (e.g., Gray et al. 2015). Indeed, organizations' physical designs that limit interactions between individuals are found to hinder collaborative innovation efforts (e.g., Catalini 2018) or to prevent learning from others (e.g., Lee 2019). Somewhat surprisingly, office designs such as open-space (compared with closed-space) offices are found to decrease face-to-face interactions and to increase electronic communication among individuals, presumably because of privacy concerns (Bernstein and Turban 2018). Szulanski and Lee (2020) note that some types of knowledge (e.g., tacit knowledge) can be best transferred through face-to-face interactions, and therefore, these new forms of organizational physical space may have substantive effects on learning. In sum, it would be worthwhile to gain a better

understanding of the effects of organizational spatial design on organizational learning.

4. It would also be worthwhile to understand the effects of sharing a social identity on organizational learning. Individuals classify themselves and others into categories based on common characteristics, such as demographic characteristics, organizational affiliations, or national identities (Hogg and Turner 1985, Ashforth and Mael 1989). A shared social identity exists when individuals perceive themselves as belonging to the same group and the group is important to them. This shared identity can exert a powerful influence on what group members think and feel and how they behave (Doosje et al. 1995). For example, individuals who share a social identity perceive each other more positively (Hewstone et al. 2002) and are more willing to cooperate (Tyler and Blader 2000) than individuals who do not share an identity. In the area of organizational learning, evidence exists that sharing a social identity increases the likelihood of knowledge transfer (Kane et al. 2005). Does sharing a social identity affect the interpretation of experience, the search for knowledge, or its retention? Understanding the effects of a shared identity on organizational learning processes is a fruitful area for future research.

5. A promising development in research on organizational learning is understanding how to overcome biases or challenges that occur in organizational learning, such as "superstitious learning," by which individuals draw inappropriate inferences from experience (Levitt and March 1988) or the hot-stove effect, by which individuals under-sample alternatives for which the initial experience was negative (Denrell and March 2001). Researchers are investigating the effectiveness of various organizational designs and procedures for counteracting these biases (Puranam and Maciejovsky 2020). For example, Paulus and Kenworthy (2020) discuss how to overcome biases in creating knowledge. Hinsz et al. (2020) present ways to lessen biases in retaining knowledge. Larson and Egan (2020) present procedures for overcoming biases in knowledge transfer. Understanding biases and how to reduce them provides a more unified treatment of the conditions under which experience is a good or bad teacher in organizations and, thus, would advance our understanding of organizational learning.

6. New forms of organizational structures are emerging that have implications for organizational learning. For example, hierarchies have become flatter, and self-managed and fluid teams are being used widely (Robertson 2015, Bernstein et al. 2016, Puranam 2018). What implications do these changes in structural designs of organizations have for organizational learning? Hierarchies are found to have positive influences on organizational learning in certain conditions (e.g., Bunderson and Boumgarden 2010, Bresman and Zellmer-Bruhn 2013) but are found to have negative influences on organizational learning in others (e.g., Sorenson 2003, Bresman and Zellmer-Bruhn 2013). Understanding the conditions under which hierarchy facilitates or constrains organizational learning is an important topic that would benefit from additional research.

7. The increase in platforms, especially two-sided ones, has interesting implications for organizational learning. Platforms are business models that create value by facilitating exchanges between suppliers and customers (Zhu and Liu 2018). Examples of two-sided platforms include Amazon.com or Hotels.com. Platforms are an interesting ecology for studying learning because the organizations on the platform are learning from both their own experience and the experience of other organizations on the platform. Supplier firms on these platforms, such as product manufacturers or hotels, expose and sell their products and services to a mass body of customers through the platform. Customers not only purchase products and services on these platforms, but also often leave feedback and rate supplier organizations. These attributes of platforms have interesting implications for the learning of supplier firms on the platform. Importantly, supplier firms have increased opportunities to vicariously learn from the experience of other firms and adjust their offerings or develop new products and services based on the feedback that others are receiving. In addition, operators of two-sided platforms (e.g., Amazon) can also learn from data that is accumulated on the platform and expand into businesses that they otherwise would not have (e.g., their own brand of clothes or everyday essentials such as vitamins). In other words, the cost associated with searching for new solutions is likely to significantly decrease for the players and the operator of the platform and, thus, make vicarious learning easier. At the same time, because learning becomes relatively easy on platforms, competition between supplier firms or between supplier firms and the platform operator can increase over time. Accordingly, it would be interesting to examine the dynamics of learning and competitive behavior on these platforms. It would be interesting to examine the temporal patterns of learning on these platforms, especially related to firms' attempts to prevent external knowledge transfer to competitors in the long run (e.g., Argote and Ingram 2000).

8. Another promising research agenda centers on the relationship between artificial intelligence and human intelligence in organizational learning. What role will machine learning play in organizational learning? What are the conditions under which machine learning is likely to be adopted in organizations? What are the conditions under which results of

machine learning will be trusted in organizations? How will information provided by machine-learning tools and information provided by members be integrated into the process of organizational learning?

9. Understanding the role of "big data" in organizational learning also merits attention. From the search of sites by individuals to locations of phone calls, large amounts of data are currently available to organizations. Will greater availability of data enable organizational learning? As Ocasio et al. (2020) note, information has to be attended to in order to have an effect, so factors affecting attention come into play in selecting and interpreting data. For high-level strategic decisions in organizations, there is often a paucity of data rather than an abundance of it (e.g., March et al. 1991). What data will organization members attend to? How will information based on big data be combined with more qualitative judgments?

10. Career trajectories have changed for individuals, and individuals now actively search for better jobs and engage in voluntary turnover instead of staying in a single organization throughout their entire careers (e.g., Cappelli 1999, Direnzo and Greenhaus 2011). Because of this phenomenon, possessing valuable and exclusive knowledge has become increasingly important for organizational members. What implications does this have for knowledge creation, transfer, and retention in organizations? Will individuals with more valuable knowledge be less willing to apply their knowledge or share their knowledge with others in an organization? Will any of an individual's knowledge be retained by the organization if the individual were to leave? How would this phenomenon affect organizations' motivation to invest in firm-specific human capital (e.g., Becker 1962) and subsequent knowledge creation processes in organizations?

11. Another interesting phenomenon related to organizational learning is that organizational knowledge that creates competitive advantage might become obsolete quicker than before because of fast-changing environments. What implications does this have for organizational learning? For example, when does prior experience facilitate organizational learning (e.g., Cohen and Levinthal 1990), and when does it act as inertia and hinder organizational learning? What kind of knowledge should be retained, and what knowledge should be purged to sustain competitive advantage? Researchers have argued that knowledge depreciation or "organizational forgetting" can reduce inertia and enable adaptation (Jain 2020) and improvisation (Miner and O'Toole 2020) in organizations. Understanding the conditions under which knowledge depreciation is a source of competitive advantage is an important agenda for future research.

12. How should organizations think about hiring or firing employees in these fast-changing environments? On the one hand, valuable prior knowledge is embedded in employees. On the other hand, new knowledge stock cannot be easily acquired in a short period of time by the same individuals (e.g., Dierickx and Cool 1989). When is it better to hire individuals who have the desired knowledge, and when is it better to develop the knowledge internally in current employees? In sum, more research on organizational learning in quickly changing environments is needed.

13. On the methodological side, developments in neuroscience have the potential to advance our understanding of organizational learning (Senior et al. 2011). Using these tools has led to advances in our understanding of individual learning. For example, Laureiro-Martínez et al. (2015) use functional magnetic resonance imaging (fMRI) to study the exploitation versus exploration decisions of individuals. Because individuals are the mechanism through which much of organizational learning occurs, these studies have implications for organizational learning. Currently, studies using functional magnetic resonance imaging typically confine one individual inside an fMRI machine and ask the individual to respond to stimuli presented on a screen. Shamay-Tsoory and Mendelsohn (2019) enumerate the limitations of tools such as fMRI for studying social cognition and memory: individual participants are not able to act on or influence the environment, participants' movements are limited, and participants' cognitive abilities are assessed on tasks that do not capture the richness of real-life experience. These features would also limit our understanding of organizational learning. To address them, Shamay-Tsoory and Mendelsohn (2019) advocate using portable technology, such as portable electroencephalography and functional near-infrared spectroscopy, to study behavior. Portable neuroscience tools have been used to study group learning (see Håkonsson et al. 2016). Their development holds promise for understanding the processes of organizational learning.

14. Another fruitful trend is using laboratory or field experiments to advance the understanding of organizational learning. Laboratory or field experiments have the advantage of enabling the establishment of causality and permitting the investigation of mechanisms' underlying effects. Examples of studies that have used laboratory experiments to study the topic of organizational learning include Cohen and Bacdayan (1994), Fang (2012), Kane et al. (2005), Liang et al. (1995), and Schilling et al. (2003). Field experiments on the topic have been relatively scant but are growing in number. Some examples include Bernstein (2012), Di Stefano et al. (2014), and Song et al. (2018). Compared with field experiments, laboratory experiments allow more control over extraneous variables. However, because of the controlled setting,

results from a laboratory experiment are typically lower in external validity than field experiments. Field experiments, on the contrary, are conducted in the real-life environments of participants; thus, the findings derived from them usually have high external validity. The use of laboratory and field experiments in conjunction with empirical studies using archival data, qualitative studies, or simulations would be a powerful combination that holds promise for advancing the understanding of organizational learning.

15. We hope that the trend of investigating the mechanism through which learning occurs continues. Learning is determined by the opportunity to learn, the motivation to learn, and the ability to learn (Argote et al. 2003, Dahlin et al. 2018). It would be useful to use this framework to understand better what promotes and hinders organizational learning in general. For example, opportunities to learn could depend on social networks (e.g., Reagans and McEvily 2003, Lovejoy and Sinha 2010). The motivation to learn could depend on incentives (e.g., Lee and Meyer-Doyle 2017, Lee and Puranam 2017, Clark et al. 2018). The ability to learn could depend on prior experience (e.g., Cohen and Levinthal 1990) and training (see Bell et al. 2017 for a review). In sum, a deeper understanding of the mechanisms through which learning is promoted or hindered would be valuable.

Conclusion

Organizational learning is a vibrant research area that has attracted researchers from a variety of disciplines. From understanding the microfoundations of organizational learning to determining its implications for the strategic behavior of firms, researchers are advancing knowledge. In this article, we aimed to take stock of what is known about organizational learning, identify important gaps in our knowledge, and suggest future directions that are likely to lead to greater understanding. These future directions include discerning how important technological and organizational developments, such as machine learning and new organizational forms, affect organizational learning. We hope that our work stimulates additional research on organizational learning. Not only will further research on organizational learning advance theory, but it also has the potential to advance practice. Learning is a key contributor to the performance of organizations and a major source of their competitive advantage.

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